Math 55 quiz 03/17/2026

You must type or neatly handwrite each answer on a separate piece of paper and select the pages for each question when you upload to Gradescope. Feel free to use more than one sheet for an answer if needed. Note that both PDF and picture/photo format (JPG) are accepted. You must show your work. During the quiz, you may consult the textbook, your own notes and other course materials you yourself prepared before the beginning of the quiz. You may not consult resources online. You can post a private question to Ed Discussion to instructors only, but do not make public posts on Ed Discussion, Discord, or other online forums.

By submitting this quiz, you agree to adhere to the statement below: "As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others. All of the work submitted here is mine, and I have not discussed the content of this exam with any other person. I have not consulted any resources beyond the course materials and my notes during the exam. I have not used calculators, computers, or the internet to do calculations or look up answers."

1. Prove that every connected simple graph which is not itself a tree must have at least three different spanning trees.

2. Define the "n-dimensional hypercube" graph $Q_n = (V_n, E_n)$ as

   $V_n = \{\text{"Binary sequences of n elements," } v = (v_1, ..., v_n) | v_i \in \{0, 1\}\}$
   $E_n = \{\{a, b\} \subseteq V | a \text{ and } b \text{ differ in exactly one bit, i.e. there exists a unique } i \in \{1, ..., n\} \text{ such that } a_i \neq b_i\}$.

   Let S_n be the set of all spanning trees of Q_n. We have shown in a homework that Q_n is connected for each $n \geq 1$ and since every connected graph has a spanning tree we must have that S_n is a non-empty set. It follows that $C_n = \{\text{"The number of leaves in the tree T"} : T \in S_n\}$ is a non-empty set of integers. Furthermore this set is bounded above by $c \leq |V_n|$ for all $c \in C_n$ and for each n and so each C_n will always have a unique maximal element.

   Define the function $f : \mathbb{N} \to \mathbb{N}$ by letting $f(n)$ equal the unique maximal element of C_n for the graph Q_n as defined above.

   Prove or disprove that $f(n) \geq 2^{n-1}$ for every $n \in \mathbb{N}$.